Badvel Veera Sai Chetan Reddy

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Summary

Computer Science graduate specializing in Artificial Intelligence with hands-on experience building AI, machine learning, backend, and data analytics applications. Proficient in Python, FastAPI, SQL, machine learning, LLMs, Retrieval-Augmented Generation (RAG), and ETL pipelines. Passionate about developing scalable, data-driven software solutions and applying AI to solve real-world problems.

Skills

Languages	Python, Java, SQL
Machine Learning	Scikit-learn, Random Forest, Decision Trees, Linear/Logistic Regression, SVM, KMeans, PCA, Time-Series Forecasting (ARIMA, Prophet, Statsmodels), Feature Engineering, Model Evaluation, Deep Learning (coursework)
AI & LLMs	LLMs, RAG, LangChain, Prompt Engineering, Semantic Search, Vector Databases (ChromaDB)
Backend & Data	FastAPI, REST APIs, Pydantic, Pandas, NumPy, ETL, PostgreSQL, Power BI, Matplotlib
Tools	Git, GitHub Actions (CI), Unit Testing, Jupyter, Gradio, Streamlit, Ollama

Projects

A Meta-Learning Framework for Automated Time-Series Model Recommendation `GitHub`

- Built an end-to-end meta-learning framework that extracts 10 statistical meta-features (ADF statistic, lag-1 autocorrelation, trend slope) to recommend the better forecasting model for unseen time series.
- Compared ARIMA(1,1,1) and Prophet on 1,359 M4 series (500 Monthly, 500 Yearly, 359 Weekly) using RMSE; trained a Random Forest meta-classifier with 69.12% holdout accuracy and confidence scores, and built a Streamlit app for batch CSV inference.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Statsmodels, Prophet, Streamlit

PokéDex AI: RAG Question Answering & Recommendation System `GitHub`

- Built a local Retrieval-Augmented Generation (RAG) system using FastAPI, ChromaDB, Sentence Transformers, and Phi-3 (Ollama) over 1,576 documents, with query parsing, metadata filtering, semantic-search fallback, and context-grounded prompting.
- Developed a recommender, team builder, and team analyzer with LLM explanations and rule-based fallback, served via REST APIs and a 4-tab Gradio interface.
- **Tech Stack:** Python, FastAPI, ChromaDB, Sentence Transformers, Ollama, Phi-3, LangChain, Gradio, Pydantic

Enterprise SQL Analytics and Business Intelligence Platform `GitHub`

- Built a Python (Pandas) ETL pipeline that streams a 5.67 GB e-commerce CSV in chunks, cleans and validates a 309,967-record sample, and loads a normalized 4-table PostgreSQL data model.
- Authored 39 SQL analytical queries (window functions, funnel and customer analysis) and a 4-page Power BI dashboard (48 visuals); found smartphones drive 73.4% of sample revenue.
- **Tech Stack:** Python, Pandas, PostgreSQL, SQL, Power BI, ETL, Data Validation

Education

- **Manipal Institute of Technology** — Bengaluru, India
  Bachelor of Technology - Computer Science (Artificial Intelligence) — *Sept 2022 – June 2026*

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing

Certifications

- IBM Data Science Professional Certificate
- Google Data Analytics Professional Certificate
- AWS Data Engineering (Foundations)
- Java Programming: Solving Problems with Software